

ACCOUNT AGE VS THE TONE OF VOICE

Introduction

Online platforms are constantly battling toxic content like harassment and hate speech. While many factors can lead to such behavior, our project investigates one specific question: **Are users with new accounts more likely to post toxic comments?** Understanding this relationship can help platforms develop better tools to identify and mitigate harmful behavior early, creating a safer and more welcoming online environment for everyone.

Objectives

- How does the **age of a user's account** correlate with the likelihood of them posting a toxic comment?
- Do certain **topics** have a significantly higher proportion of toxic comments than others?

Data Collection

- Used the **Reddit API (PRAW)** to collect comments and metadata from selected subreddits.
- Retrieved **key fields**: comment text, subreddit topic, user account age, and scores.
- Applied Python tools (**PRAW, pandas**) to clean, structure, and store the data.
- Focused on gathering high-quality, contextual data to support **toxicity analysis**.

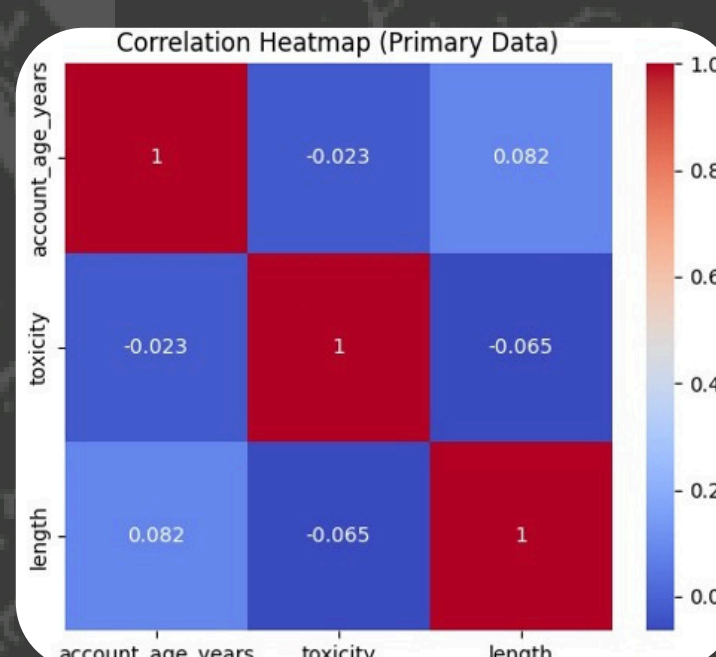
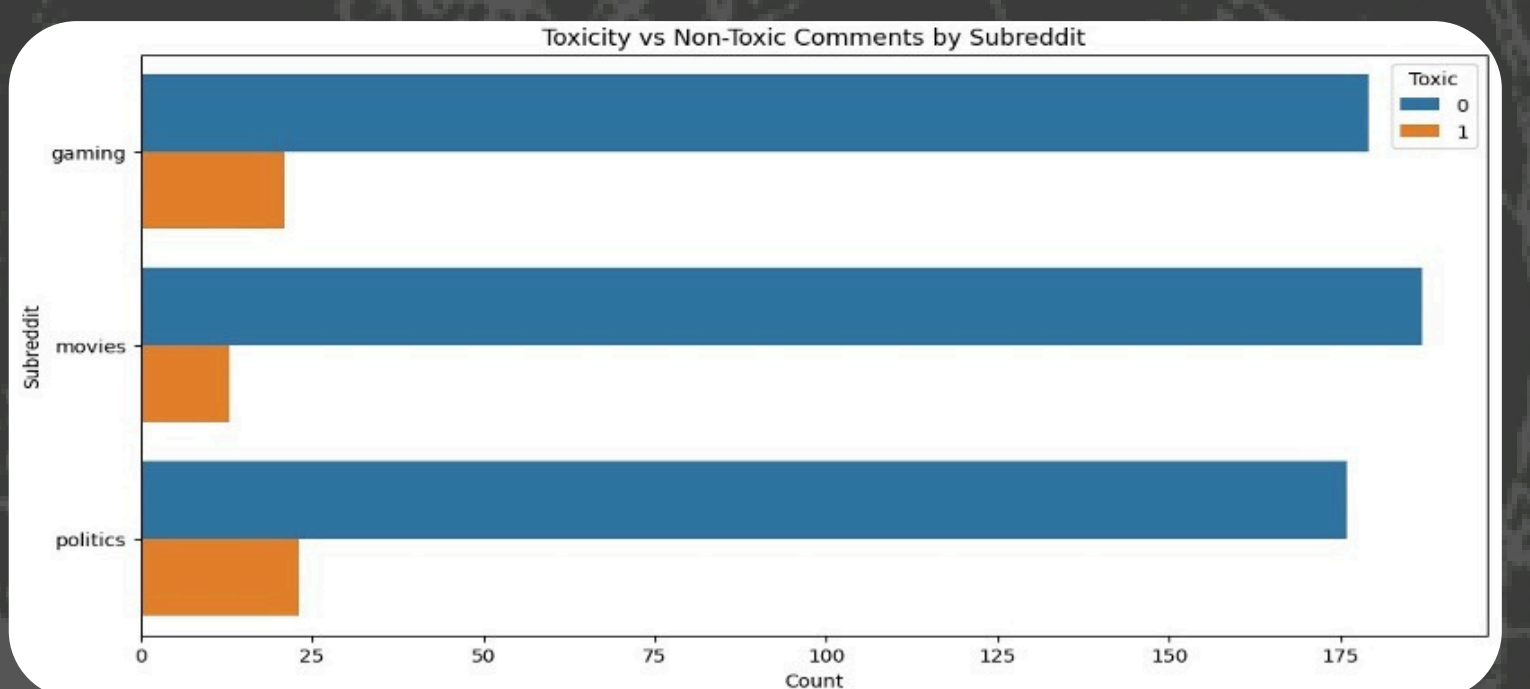
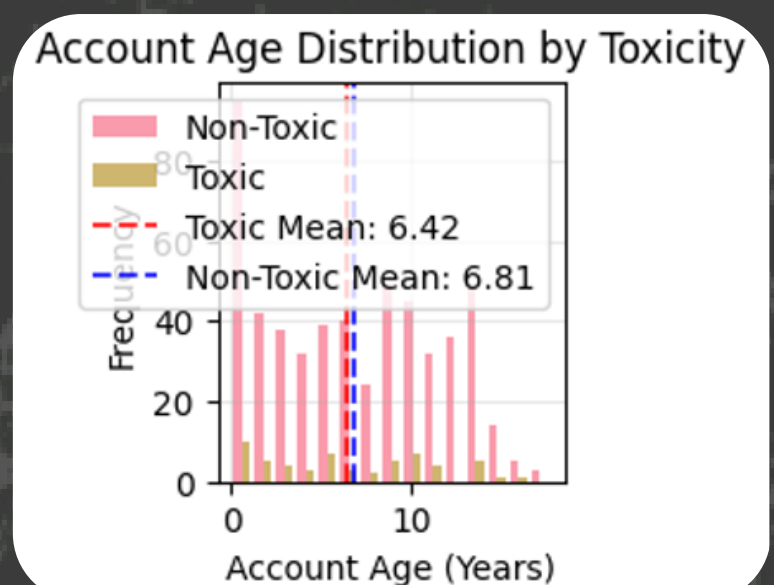
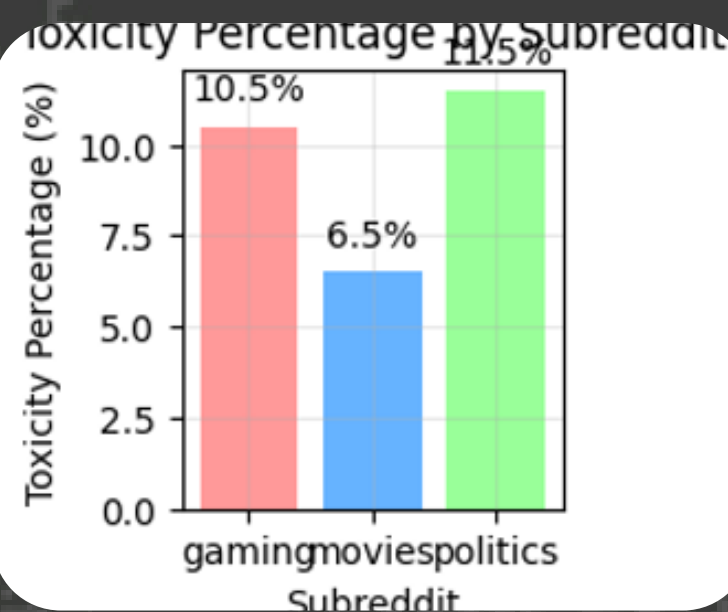
Data Analysis

Our exploratory analysis revealed critical patterns:

- **Toxicity is Rare:** The vast majority of comments (over 90%) are non-toxic.
- **Account Age is a Weak Clue:** Accounts with toxic comments tend to be a bit newer than non-toxic ones.
- **Topic Matters Most:** The community topic is a major factor in toxicity.
 - **r/politics** had the highest toxicity rate (~10.5%).
 - **r/gaming** had a moderate rate (~6.5%).
 - **r/movies** had the lowest rate (~2.5%).

Models And Findings

- We tested three models: a **Baseline Linear Regression**, **Logistic Regression + TF-IDF**, and **Linear SVM + TF-IDF**.
- The Baseline model showed **high accuracy** but failed to detect toxic comments, revealing **strong class imbalance**.
- **Logistic Regression + TF-IDF** delivered the **best overall performance**, with **strong accuracy** and improved **toxic-class detection**.
- **Linear SVM + TF-IDF** performed well on non-toxic comments but struggled with toxic recall compared to Logistic Regression.
- Overall, **TF-IDF-based models** significantly **outperformed** the baseline, proving that text representation is critical for **toxicity classification**.



Conclusion

Key Insights & Implications

- **Context Matters Most:** Toxicity is driven more by the topic of discussion than by user characteristics.
- **Account Age Is Not a Strong Predictor:** Newer accounts are not inherently more toxic.
 - Policies that over-target new users risk being both ineffective and unfair to legitimate newcomers.
- **Future Direction:** Effective toxicity detection must move beyond basic metadata. Incorporating **NLP-based text analysis** and community context will be essential for building more accurate, fair, and reliable models.